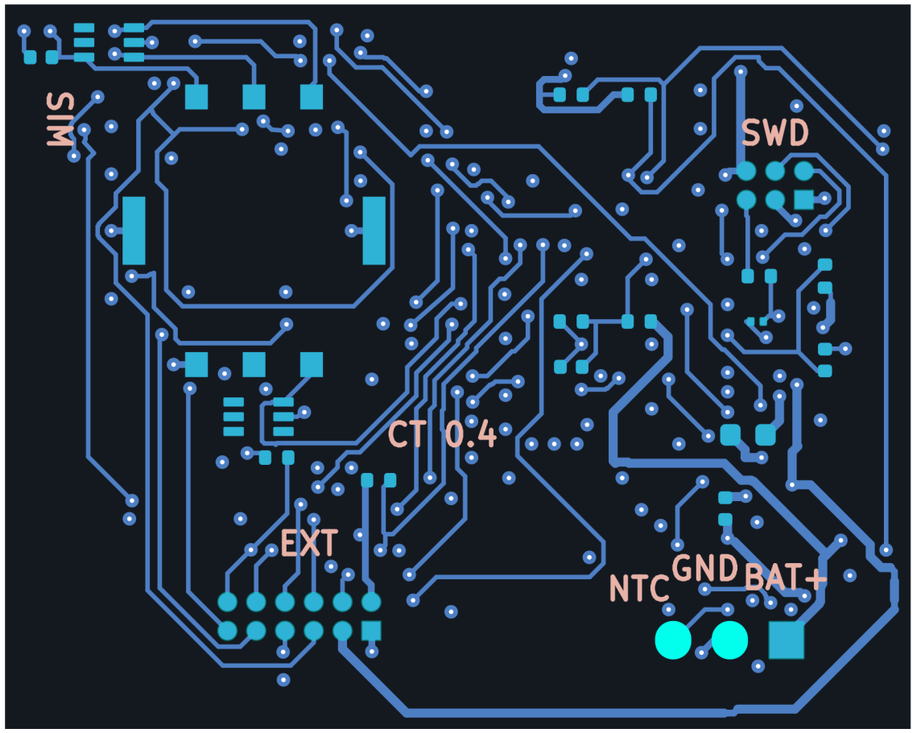
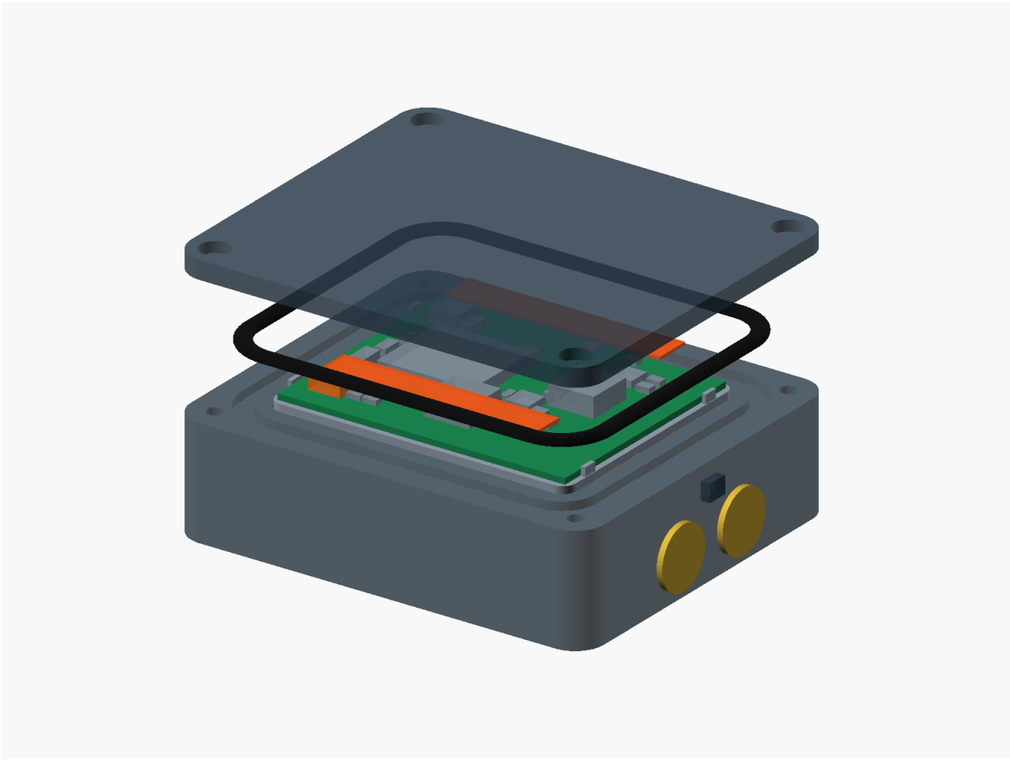
COLLAR TRACKER / DESIGN REVIEW

r04i PCB verified • swim-r02 populated-fit prototype • final part selections and physical tests remain open

40 × 32 mm / FOUR-LAYER PCB



SEALED HOUSING / EXPLODED MODEL



PCB CHECKS COMPLETE

Native DRC / ERC / connectivity / parity: zero findings

Two real JLC records: 0 Danger / 0 Warning each

Identical ZIP, unchanged rules, 29 analyzed rows each

82 local GND pads; two continuous inner GND layers

93 / 93 signal-via return screens; SIM lane clear

SMT assembly, power and RF qualification remain open

MECHANICAL PROTOTYPE

Rigid body: 57.4 × 49.4 × 17.6; 58.7 including the key

With collar boot: 60.4 × 52.4 × 22.4

Continuous lid O-ring; two independent contact seals

Target: 2 m fresh water / 60 min — provisional conditions

49 component heights checked; pack and antennas still open

Carrier reliefs and M3 contact retention; no physical IP test